

Single-Peak Crow–Kimura Dynamics: Complete Finite-Genome Spectrum and Exact Projective Gap

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Abstract

For the continuous-time mutation–selection equation on binary strings of length L , with symmetric point mutation and one master-sequence fitness spike, we identify the nonlinear simplex flow exactly with a projectivized positive linear semigroup. Walsh decomposition shows that each Hamming eigenvalue retains an explicit codimension-one multiplicity, while all remaining eigenvalues are the roots of one degree- $(L + 1)$ secular polynomial.

1 Frozen mutation–selection owner

Let $\mathcal{H}_L = \mathbb{R}^{\{0,1\}^L}$, let F_i flip bit i , and let e_0 denote the all-zero genotype. For $L \geq 1$ and $U, s > 0$ define

$$M_L = \frac{U}{L} \sum_{i=1}^L (F_i - I), \quad A_L = M_L + s e_0 e_0^\top. \quad (1)$$

Both matrices are real symmetric and $1^\top M_L = 0$. On the probability simplex the Crow–Kimura equation in this normalization is

$$\dot{p} = A_L p - (1^\top A_L p) p = A_L p - s p_0 p. \quad (2)$$

Theorem 1 (Exact projectivization and global limit). *For every probability vector p^{in} ,*

$$p(t) = \frac{e^{tA_L} p^{\text{in}}}{1^\top e^{tA_L} p^{\text{in}}} \quad (3)$$

is the unique solution of (2). The denominator is positive, the simplex is preserved, and $p(t)$ converges to the uniquely normalized positive Perron eigenvector of A_L .

Proof. Write $q(t) = e^{tA_L} p^{\text{in}}$ and $Z(t) = 1^\top q(t)$. Positivity of the irreducible hypercube semigroup gives $q(t) > 0$ for $t > 0$, hence $Z(t) > 0$. Moreover $Z' = 1^\top A_L q = s q_0$. Differentiating q/Z gives exactly (2). The same calculation shows that its entries sum to one. Since A_L is irreducible Metzler and symmetric, its largest eigenvalue is simple and has a strictly positive eigenvector. The Perron coefficient of every nonzero nonnegative initial vector is positive; dividing the spectral expansion by $Z(t)$ proves the limit. $\square$

The phrase *projective semigroup owner* records the theorem content of the original manuscript round.

2 Complete finite-genome spectrum

For $S \subseteq \{1, \dots, L\}$ let

$$\phi_S(x) = 2^{-L/2} (-1)^{\sum_{i \in S} x_i}.$$

These Walsh characters form an orthonormal basis and

$$M_L \phi_S = d_{|S|} \phi_S, \quad d_k = -\frac{2Uk}{L}, \quad \langle e_0, \phi_S \rangle = 2^{-L/2}. \quad (4)$$

Put $m_k = L!/[k!(L-k)!]$, $w_k = m_k/2^L$.

Theorem 2 (Full multiplicity and secular factorization). *The complete characteristic polynomial of A_L is*

$$\det(\lambda I - A_L) = q_L(\lambda) \prod_{k=0}^L (\lambda - d_k)^{m_k - 1}, \quad (5)$$

$$q_L(\lambda) = \prod_{k=0}^L (\lambda - d_k) - s \sum_{k=0}^L w_k \prod_{j \neq k} (\lambda - d_j). \quad (6)$$

Thus d_k is retained with multiplicity $m_k - 1$ and the remaining $L + 1$ eigenvalues are precisely the roots, away from the poles, of

$$1 = s \sum_{k=0}^L \frac{w_k}{\lambda - d_k}. \quad (7)$$

The multiplicities in (5) sum to 2^L .

Proof. Let $E_k = \text{span}\{\phi_S : |S| = k\}$. The subspace of E_k whose Walsh coefficients sum to zero is orthogonal to e_0 and has dimension $m_k - 1$; A_L acts there by d_k . Its orthogonal complement over all k has basis

$$u_k = m_k^{-1/2} \sum_{|S|=k} \phi_S, \quad \langle e_0, u_k \rangle = \sqrt{w_k}.$$

The restriction of A_L is $D + s v v^\top$, where $D = \text{diag}(d_0, \dots, d_L)$ and $v_k = \sqrt{w_k}$. The matrix determinant lemma gives (6); adjoining the retained orthogonal spaces gives (5). $\square$